

# Vivado 2019.1 - IP Import

## Import RTL IP

### [Creating a custom IP block in Vivado - FPGA Developer](#)

Create and Package New IP

**Peripheral Details**  
Specify name, version and description for the new peripheral

Name:

Version:

Display name:

Description:

IP location:  ...

☐ Overwrite existing

? < Back Next > Finish Cancel

Create and Package New IP

**Add Interfaces**  
Add AXI4 interfaces supported by your peripheral

☐ Enable Interrupt Support

Interfaces

- S00\_AXI

matrix\_multiplication\_v1.0

S00\_AXI

Name:

Interface Type:

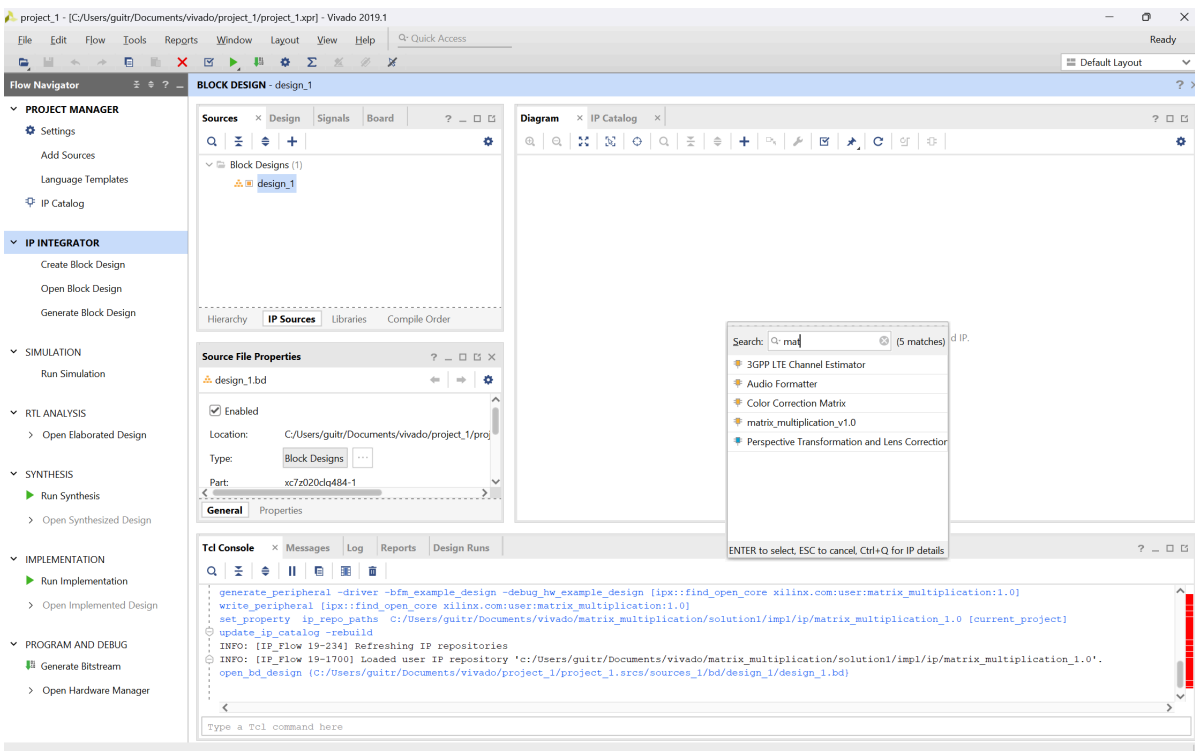
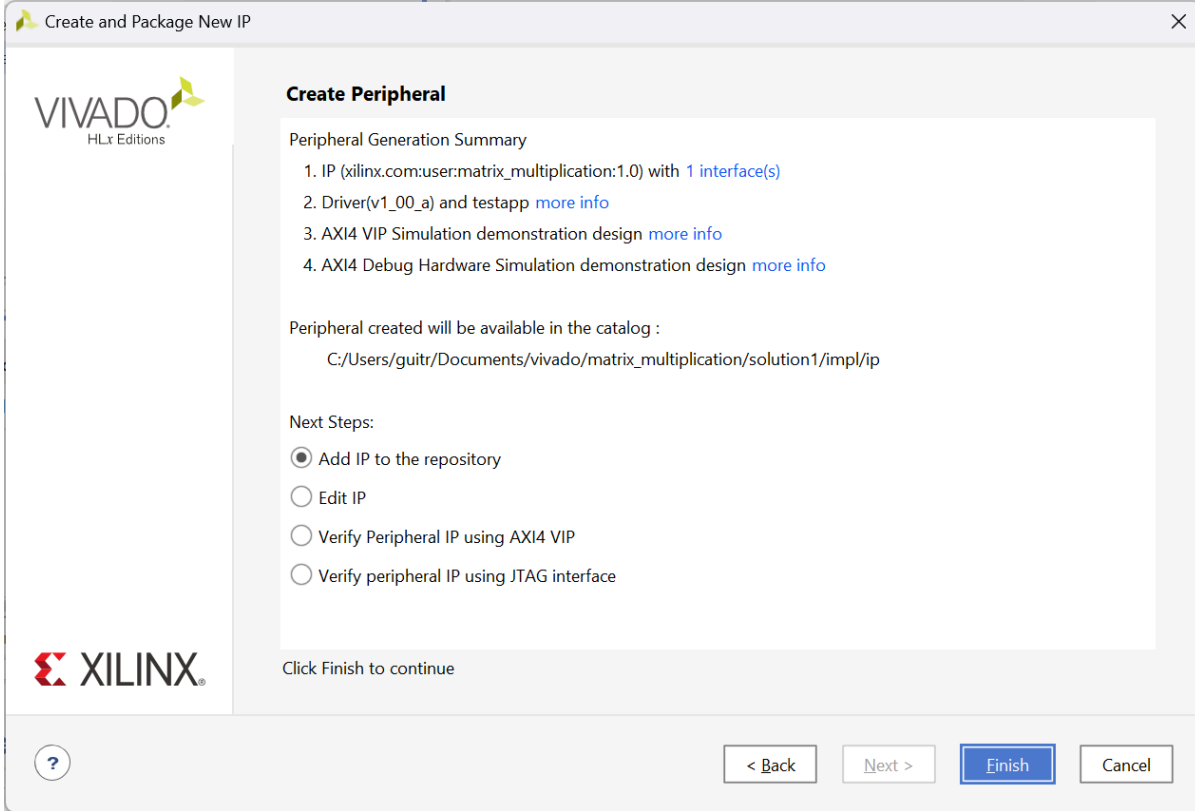
Interface Mode:

Data Width (Bits):

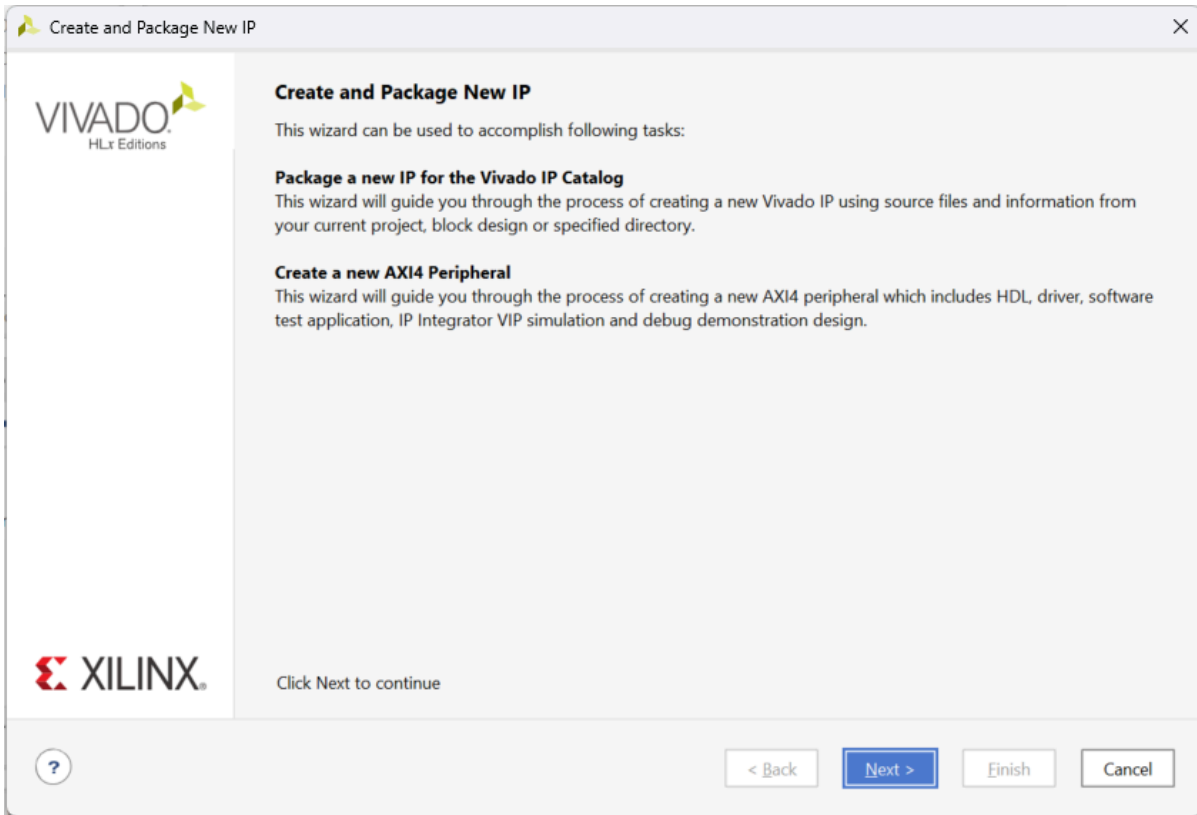
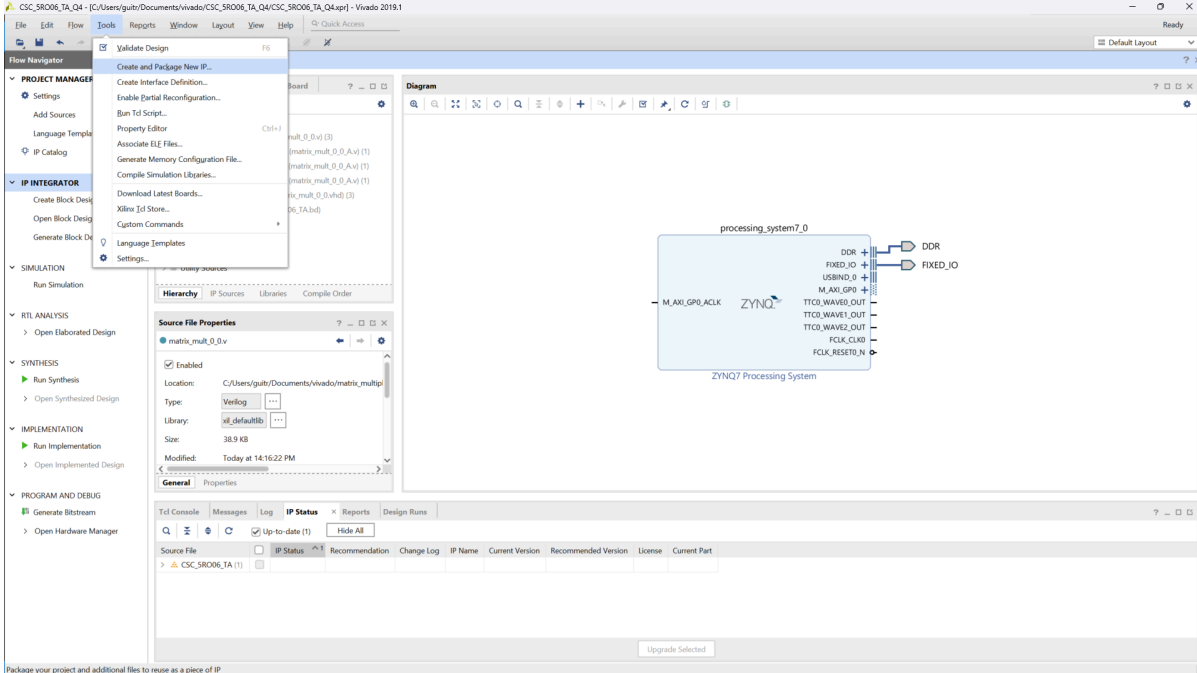
Memory Size (Bytes):

Number of Registers:  [4..512]

? < Back Next > Finish Cancel



## Create RTL IP



Create and Package New IP

Create Peripheral, Package IP or Package a Block Design

Please select one of the following tasks.

Packaging Options

☐ Package your current project  
Use the project as the source for creating a new IP Definition.

☐ Package a block design from the current project  
Choose a block design as the source for creating a new IP Definition.  
Select a block design: CSC\_5R006\_TA

☒ Package a specified directory  
Choose a directory as the source for creating a new IP Definition.

Create AXI4 Peripheral

☐ Create a new AXI4 peripheral  
Create an AXI4 IP, driver, software test application, IP Integrator AXI4 VIP simulation and debug demonstration design.

?

< Back

Next >

Finish

Cancel

Create and Package New IP

Package a Specified Directory

Select the directory where sources to be packaged are located.

Directory: C:/Users/guitr/Documents/vivado/matrix\_multiplication/matrix\_multiplication/solution\_0\_0\_8/impl/ip

☐ Package as a library core

?

< Back

Next >

Finish

Cancel

Confirm Copy Sources

?

The project source files will be copied to the indicated IP Location directory and an editing project will be opened. Would you like to continue and copy?

OK

Cancel

Create and Package New IP

### Edit in IP Packager Project Name

Enter a name for your project and specify a directory where the project data files will be stored

Project name:

Project location:  ...

Edit IP project will be created at: C:/.../vivado/CSC\_5RO06\_TA\_Q4/CSC\_5RO06\_TA\_Q4.tmp/CSC\_5RO06\_TA\_IP\_Package

? < Back Next > Finish Cancel

Create and Package New IP



### New IP Creation

The following pieces of information will be gathered:

- Identification information based on top module name
- Family compatibility based on part in the project
- File(s) from Synthesis and Simulation file sets
- Ports from the file containing the top module
- Parameters from the file containing the top module
- Bus Interfaces based on port names
- Address Spaces and Memory Maps based on inferred bus interfaces

Following file will be created on disk along with corresponding customization files:  
C:/Users/guitr/Documents/vivado/matrix\_multiplication/matrix\_multiplication/solution\_0\_0\_8/impl/ip/component

Click Finish to continue

? < Back Next > Finish Cancel

CSC\_SRO06\_TA\_IP\_Package - [c:/users/guitr/documents/vivado/csc\_sro06\_ta\_q4/csc\_sro06\_ta\_q4tmp/CSC\_SRO06\_TA\_IP\_Package.vpr] - Vivado 2019.1

File Edit Flow Tools Reports Window Layout View Help

Flow Navigator PROJECT MANAGER - CSC\_SRO06\_TA\_IP\_Package

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog
- Edit Packaged IP

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG

- Generate Bitstream
- Open Hardware Manager

Sources

- Design Sources (3)
- matrix\_mult\_0\_0 [matrix\_mult\_0\_0]
- matrix\_mult\_0\_0[behav] [matrix\_mult\_0\_0]
- IP-XACT (1)
- Constraints
- Simulation Sources (2)
- Utility Sources

Hierarchy Libraries Compile Order

Properties

Select an object to see properties

Project Summary Package IP - matrix\_mult\_0\_0

Overview Dashboard

Settings Edit

Project name: CSC\_SRO06\_TA\_IP\_Package  
Project location: c:/users/guitr/documents/vivado/csc\_sro06\_ta\_q4/csc\_sro06\_ta\_q4tmp  
Product family: Zynq 7000  
Project part: ZedBoard Zynq Evaluation and Development Kit [xc7z020dpg484-1]  
Top module name: matrix\_mult\_0\_0  
Target language: VHDL  
Simulator language: Mixed

Synthesis Implementation

Status: Not started  
Messages: No errors or warnings  
Part: xc7z020dpg484-1  
Strategy: Vivado Synthesis Defaults  
Report Strategy: Vivado Synthesis Default Reports  
Incremental synthesis: None

DRC Violations

Run Implementation to see DRC results

Timing

Run Implementation to see timing results

Utilization

Run Synthesis to see utilization results

Power

Run Implementation to see power results

Tcl Console Messages Log Reports Design Runs

| Name   | Constraints   | Status      | WNS | TNS | WHS | THS | TPWS | Total Power | Failed Routes | LUT | FF | BRAMs | URAM | DSP | Start | Elapsed | Run Strategy                                                | Report Strategy                                                    |
|--------|---------------|-------------|-----|-----|-----|-----|------|-------------|---------------|-----|----|-------|------|-----|-------|---------|-------------------------------------------------------------|--------------------------------------------------------------------|
| ip_0_1 | constraints_1 | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Synthesis Defaults (Vivado Synthesis 2019)           | Vivado Synthesis Default Reports (Vivado Synthesis 2019)           |
| impl_1 | constraints_1 | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Implementation Defaults (Vivado Implementation 2019) | Vivado Implementation Default Reports (Vivado Implementation 2019) |

CSC\_SRO06\_TA\_IP\_Package - [c:/users/guitr/documents/vivado/csc\_sro06\_ta\_q4/csc\_sro06\_ta\_q4tmp/CSC\_SRO06\_TA\_IP\_Package.vpr] - Vivado 2019.1

File Edit Flow Tools Reports Window Layout View Help

Flow Navigator PROJECT MANAGER - CSC\_SRO06\_TA\_IP\_Package

PROJECT MANAGER

- Settings
- Add Sources
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- IP Catalog
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Sources

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- matrix\_mult\_0\_0 [matrix\_mult\_0\_0]
- matrix\_mult\_0\_0[behav] [matrix\_mult\_0\_0]
- IP-XACT (1)
- Constraints
- Simulation Sources (2)
- Utility Sources

Hierarchy Libraries Compile Order

IP Bus Interface Property

Input\_stream

Name: Input\_stream  
Display name:  
Description:  
Bus type vendor: xilinx  
Bus type library: inter  
Bus type name: axis

Package IP - matrix\_mult\_0\_0

Packaging Steps

- Identification
- Compatibility
- File Groups
- Customization Parameters
- Ports and Interfaces
- Addressing and Memory
- Customization GUI

Review and Package

Review and Package

Summary

Display name: matrix\_mult\_0\_0\_v1\_0  
Description: matrix\_mult\_0\_0\_v1\_0  
Root directory: c:/Users/guitr/Documents/vivado/matrix\_multiplication/matrix\_multiplication/solution\_0\_0\_8/impl/ip

After Packaging

An archive will not be generated. Use the settings link below to change your preference  
Project will be removed after completion  
[Edit packaging settings](#)

Package IP

Tcl Console Messages Log Reports Design Runs

Warning (7) Info (5) Show All

- General Messages (2 warnings)
- Warning 56-999 Vivado Synthesis ignores library specification for Verilog or SystemVerilog files. [matrix\_mult\_0\_0.v] (1 more like this)
- Analysis Results (1 warning)
- Warning 20-7363 The current top specification, 'matrix\_mult\_0\_0' of file: c:/Users/guitr/Documents/vivado/matrix\_multiplication/matrix\_multiplication/solution\_0\_0\_8/impl/p/hdl/matrix\_mult\_0\_0\_A.vhdl, does not uniquely identify a single design element. Using 'matrix\_mult\_0\_0' (File: c:/Users/guitr/Documents/vivado/matrix\_multiplication/matrix\_multiplication/solution\_0\_0\_8/impl/p/hdl/verilog/matrix\_mult\_0\_0.v)
- IP Package (4 warnings)
- Ports and Interfaces Wizard (4 warnings)
- IP Flow 19-3158 Bus interface 'input\_stream' FREQUENCY bus parameter missing from AXI interface when interface is not associated to a clock. (1 more like this)
- IP Flow 19-3157 Bus interface 'ap\_rst\_n' bus parameter POLARITY is ACTIVE\_LOW but port 'ap\_rst\_n' is not 'resetn' - please double check the POLARITY setting.
- IP Flow 19-3153 Bus interface 'ap\_clk' ASSOCIATED\_BUSIF bus parameter is missing.

Close Project

Finished packaging 'matrix\_mult\_0\_0\_8\_v1\_0' successfully.  
Package IP Location: c:/Users/guitr/Documents/vivado/matrix\_multiplication/  
matrix\_multiplication/solution\_0\_0\_8/impl/ip

Do you want to close the project?

Yes No

Add RTL IP

File Edit Flow Tools Reports Window Layout View Help Quick Access Ready

Flow Navigator

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG

- Generate Bitstream
- Open Hardware Manager

IP Catalog

- Design
- Signals
- Board
- Platform Interfaces
- Diagram
- Address Editor
- Properties
- Selection
- Tcl Console
- Messages
- Log
- Reports
- Design Runs
- IP Status

Source File Properties

matrix\_mult\_0.v

Enabled

Location: C:/Users/guix/Documents/vivado/matrix\_mult\_0.v

Type: Verilog

Library: xil\_defaultlib

Size: 38.9 KB

Modified: Today at 14:16:22 PM

General Properties

Diagram

processing\_system7\_0

DDR

FIXED\_IO

USBIND\_0

M\_AXI\_GP0

TTC0\_WAVE0\_OUT

TTC0\_WAVE1\_OUT

TTC0\_WAVE2\_OUT

FCLK\_CLK0

FCLK\_RESET0\_N

ZYNQ

ZYNQ7 Processing System

Tcl Console

Messages

Log

IP Status

Reports

Design Runs

Report is out of date because the status of one or more IPs have changed. [Run](#)

Source File

| Source File           | IP Status      | Recommendation                                | Change Log | IP Name                 | Current Version | Recommended Version | License  | Current Part   |
|-----------------------|----------------|-----------------------------------------------|------------|-------------------------|-----------------|---------------------|----------|----------------|
| CSC_SRO06_TA (1)      |                |                                               |            |                         |                 |                     |          |                |
| /processing_system7_0 | Up-to-date (1) | No changes required <a href="#">More Info</a> |            | ZYNQ7 Processing System | 5.5 (Rev. 6)    | 5.5 (Rev. 6)        | Included | xc7z020lp484-1 |

Upgrade Selected

Browse, customize, and generate cores

CSC\_SRO06\_TA\_Q4 - [C:/Users/guix/Documents/vivado/CSC\_SRO06\_TA\_Q4/CSC\_SRO06\_TA\_Q4.apl] - Vivado 2019.1

File Edit Flow Tools Reports Window Layout View Help Quick Access Ready

Flow Navigator

BLOCK DESIGN - CSC\_SRO06\_TA\*

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog

IP INTEGRATOR

- Create Block Design
- Open Block Design
- Generate Block Design

SIMULATION

- Run Simulation

RTL ANALYSIS

- Open Elaborated Design

SYNTHESIS

- Run Synthesis
- Open Synthesized Design

IMPLEMENTATION

- Run Implementation
- Open Implemented Design

PROGRAM AND DEBUG

- Generate Bitstream
- Open Hardware Manager

Sources

- Design Sources (1)
- matrix\_mult\_0\_0 (matrix\_mult\_0\_0.v) (3)
- A\_U: matrix\_mult\_0\_0\_A (matrix\_mult\_0\_0.v) (1)
- R\_U: matrix\_mult\_0\_0\_A (matrix\_mult\_0\_0.v) (1)
- C\_U: matrix\_mult\_0\_0\_A (matrix\_mult\_0\_0.v) (1)
- matrix\_mult\_0\_0 (matrix\_mult\_0\_0.v) (3)
- CSC\_SRO06\_TA (CSC\_SRO06\_TA.tcl)
- Constraints
- Simulation Sources (1)
- sim\_1 (1)
- Utility Sources

Hierarchy

IP Sources

Libraries

Compile Order

IP Properties

matrix\_mult\_0\_0.v1\_0

Version: 1.0 (Rev. 1)

Interface: AXI4-Stream

Description: matrix\_mult\_0\_0.v1\_0

Status: Production

License: Included

Change Log: [View Change Log](#)

Vendor: Xilinx, Inc.

VIN: xilinx.com:user:matrix\_mult\_0\_0\_0\_1.0

IP Catalog

Cores

AXI4

Status

License

VIN

User Repository (c:/Users/guix/Documents/vivado/matrix\_mult\_0\_0\_0\_1/impl/ip)

UserIP

matrix\_mult\_0\_0\_0\_1\_0

AXI4-Stream

Production

Included

xilinx.com:user:matrix\_mult\_0\_0\_0\_1.0

Vivado Repository

Alliance Partners

Audio Connectivity & Processing

Automotive & Industrial

AXI Infrastructure

AXIS Infrastructure

BaseIP

Basic Elements

Communication & Networking

Debug & Verification

Details

Name: matrix\_mult\_0\_0\_0\_1.0

Version: 1.0 (Rev. 1)

Interface: AXI4-Stream

Description: matrix\_mult\_0\_0\_0\_1.0

Status: Production

License: Included

Vendor: Xilinx, Inc.

Tcl Console

Messages

Log

IP Status

Reports

Design Runs

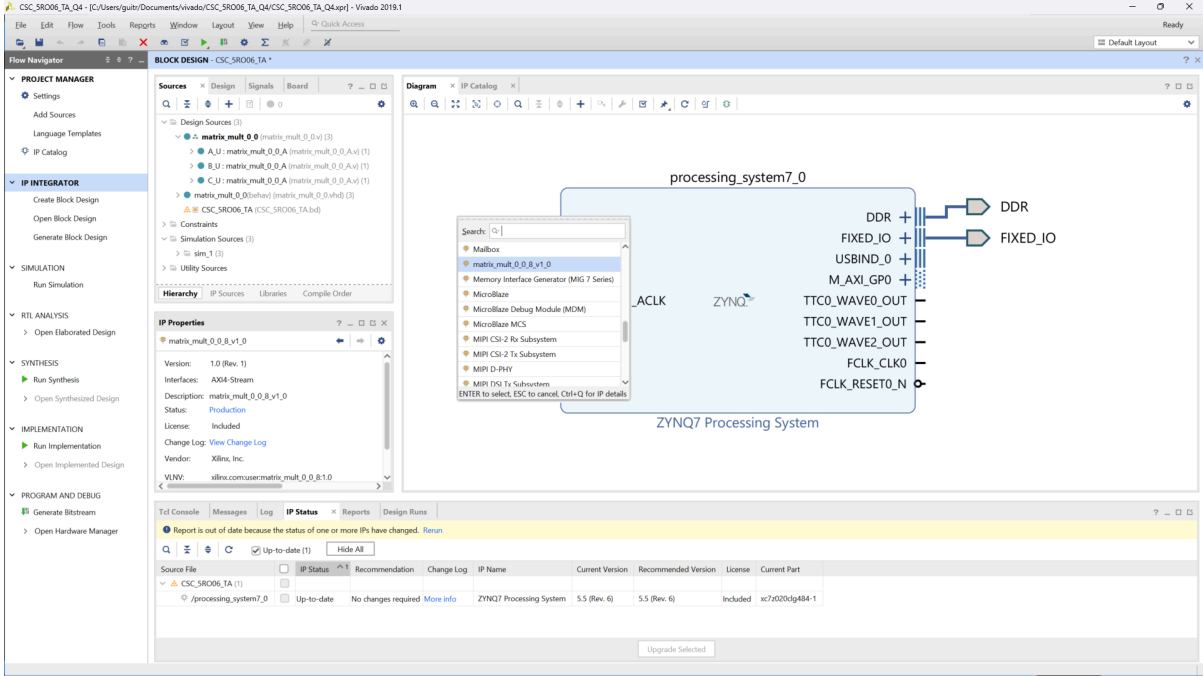
Report is out of date because the status of one or more IPs have changed. [Run](#)

Source File

| Source File           | IP Status  | Recommendation                                | Change Log | IP Name                 | Current Version | Recommended Version | License  | Current Part   |
|-----------------------|------------|-----------------------------------------------|------------|-------------------------|-----------------|---------------------|----------|----------------|
| CSC_SRO06_TA (1)      |            |                                               |            |                         |                 |                     |          |                |
| /processing_system7_0 | Up-to-date | No changes required <a href="#">More Info</a> |            | ZYNQ7 Processing System | 5.5 (Rev. 6)    | 5.5 (Rev. 6)        | Included | xc7z020lp484-1 |

Upgrade Selected

Hide disabled and incompatible IPs



By a mistery only explained by the higher universal probability, [this tutorial](#) worked.

# Import Repository



Screenshot of the Vivado 2019.1 Project Manager interface. The top menu bar includes File, Edit, Flow, Tools, Reports, Window, Layout, View, and Help. The left sidebar shows the Project Manager tree with sections: PROJECT MANAGER (Settings, Add Sources, Language Templates, IP Catalog), IP INTEGRATOR (Create Block Design, Open Block Design, Generate Block Design), SIMULATION (Run Simulation), RTL ANALYSIS (Open Elaborated Design), SYNTHESIS (Run Synthesis, Open Synthesized Design), IMPLEMENTATION (Run Implementation, Open Implemented Design), and PROGRAM AND DEBUG (Generate Bitstream, Open Hardware Manager).

The main workspace is divided into several panes:

- Sources:** Shows Design Sources (Constraints, Simulation Sources, sim\_1, Utility Sources).
- Properties:** Select an object to see properties.
- Project Summary:** Overview | Dashboard. Includes Settings (Project name: SRO06\_Q4\_0\_0\_8, Project location: C:/Users/guix/Documents/Vivado/SRO06\_Q4\_0\_0\_8, Product family: Zynq-7000, Project part: ZedBoard Zynq Evaluation and Development Kit (xc7z020clg484-1), Top module name: Not defined, Target language: VHDL, Simulator language: Mixed) and Board Part (Display name: ZedBoard Zynq Evaluation and Development Kit, Board part name: emanet.com.zedpart01.4, Board revision: d, Connectors: No connections, Repository path: C:/Xilinx/Vivado/2019.1/data/boards/board\_files, URL: http://www.zedboard.org, Board overview: ZedBoard Zynq Evaluation and Development Kit).
- Synthesis and Implementation:** Status: Not started, Messages: No errors or warnings, Part: xc7z020clg484-1.
- Design Runs:** A table showing synthesis and implementation runs.

| Name    | Constraints   | Status      | WNS | TNS | WHS | THS | TPWS | Total Power | Failed Routes | LUT | FF | BRAMs | URAM | DSP | Start | Elapsed | Run Strategy                                                | Report Strategy                                                    |
|---------|---------------|-------------|-----|-----|-----|-----|------|-------------|---------------|-----|----|-------|------|-----|-------|---------|-------------------------------------------------------------|--------------------------------------------------------------------|
| synth_1 | constraints_1 | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Synthesis Defaults (Vivado Synthesis 2019)           | Vivado Synthesis Default Reports (Vivado Synthesis 2019)           |
| impl_1  | constraints_1 | Not started |     |     |     |     |      |             |               |     |    |       |      |     |       |         | Vivado Implementation Defaults (Vivado Implementation 2019) | Vivado Implementation Default Reports (Vivado Implementation 2019) |

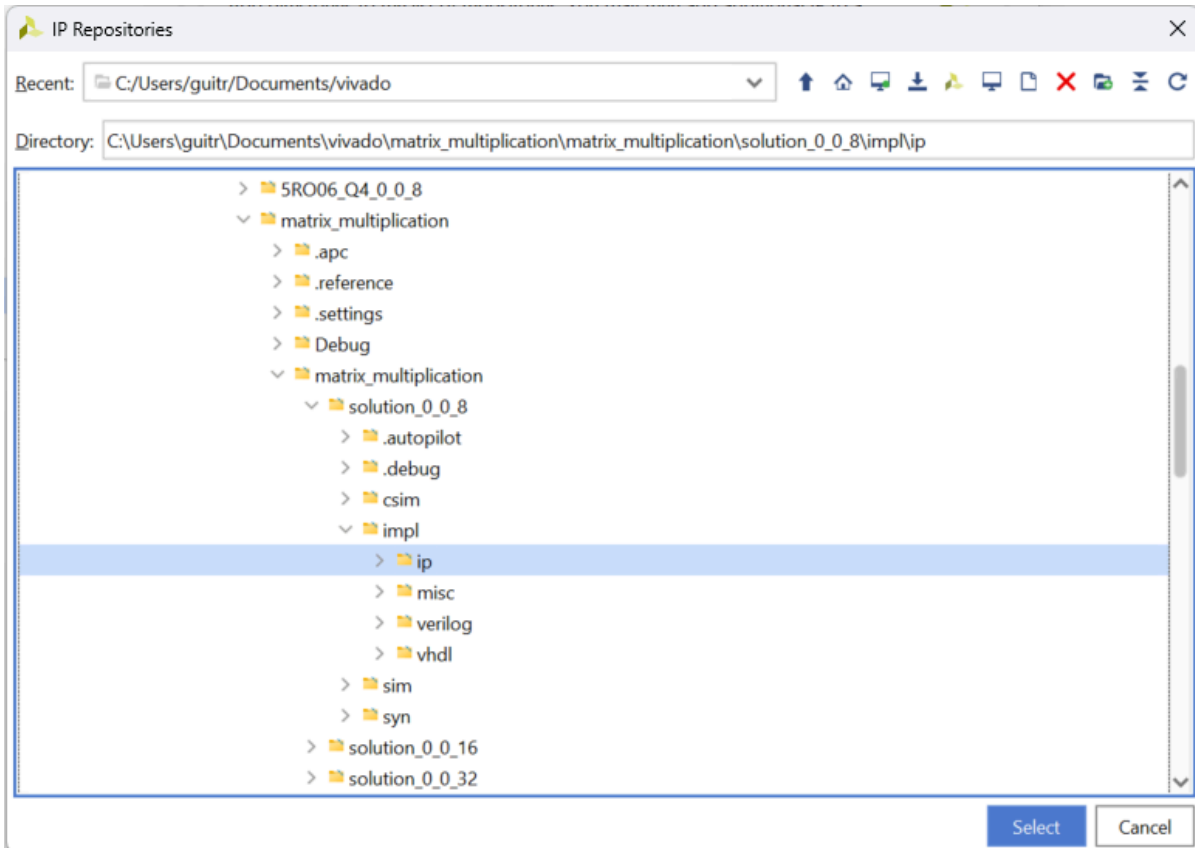
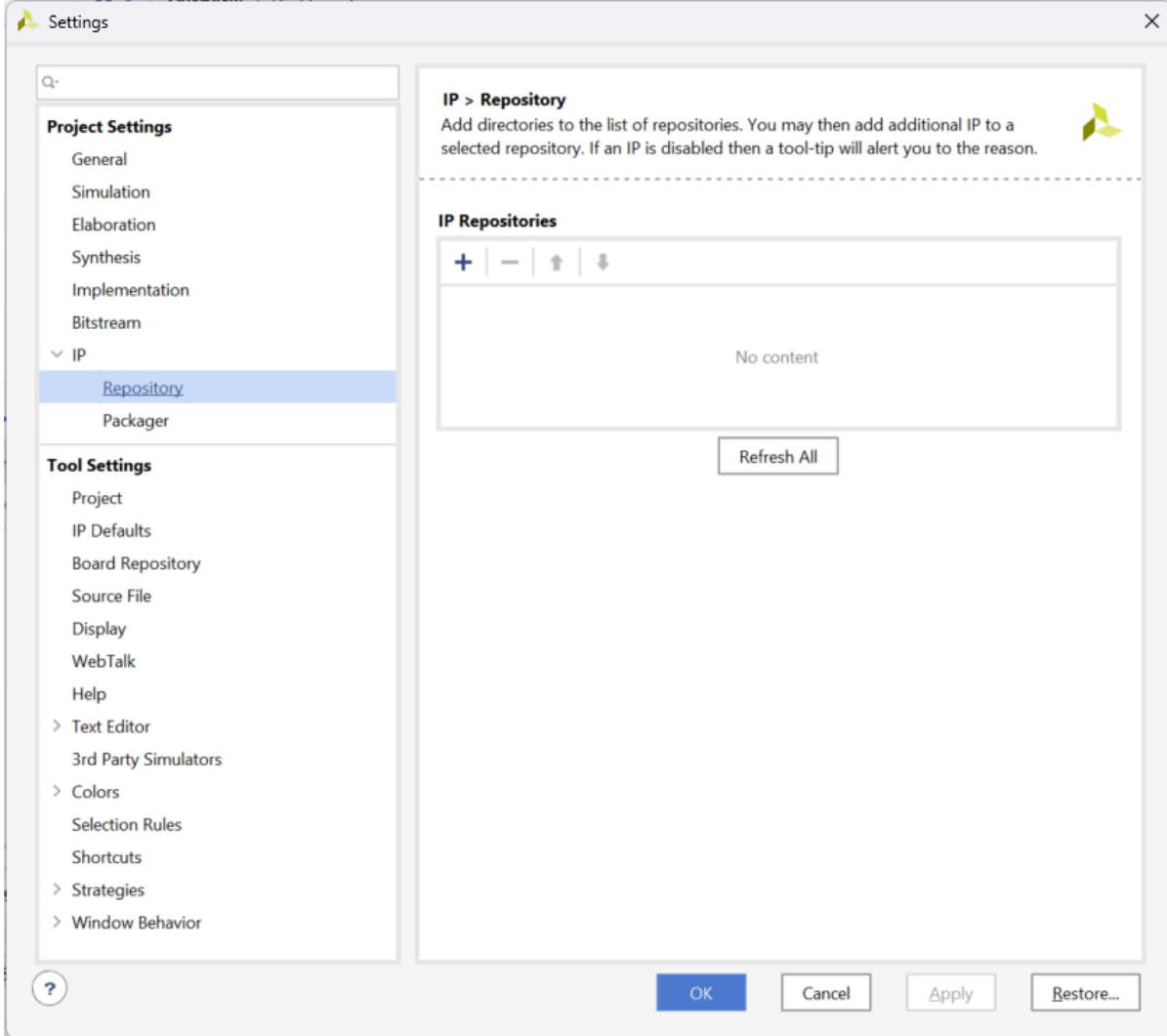
Generate outputs needed for synthesis, simulation and implementation

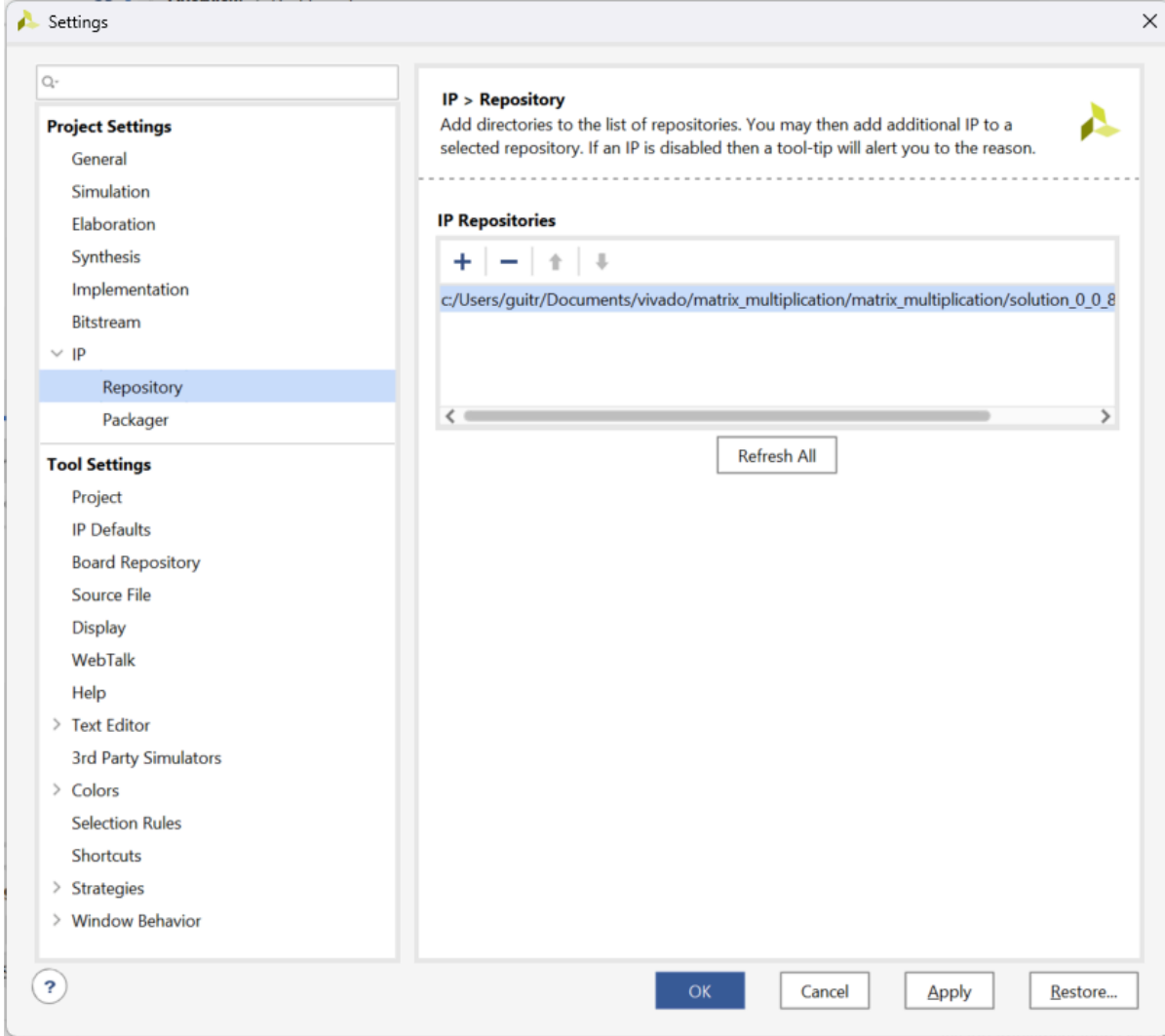
Screenshot of the Vivado Settings dialog box. The left sidebar shows the Settings tree with sections: Project Settings (General, Simulation, Elaboration, Synthesis, Implementation, Bitstream, IP) and Tool Settings (Project, IP Defaults, Board Repository, Source File, Display, WebTalk, Help, Text Editor, 3rd Party Simulators, Colors, Selection Rules, Shortcuts, Strategies, Window Behavior).

The main workspace shows the General settings for the project SRO06\_Q4\_0\_0\_8. The settings are:

- Name:** SRO06\_Q4\_0\_0\_8
- Project device:** rd Zynq Evaluation and Development Kit (xc7z020clg484-1)
- Target language:** VHDL
- Default library:** xil\_defaultlib
- Top module name:** (empty)
- Language Options:**
  - Verilog options:** verilog\_version=Verilog 2001
  - Generics/Parameters:** (empty)
  - Loop count:** 1,000

Buttons at the bottom: OK, Cancel, Apply, Restore...





[Vivado 2019.1 - Create Block Diagram](#)